

PURBANCHAL UNIVERSITY

2024

Master of Computer Application (M.C.A.)/First Semester/*Final*

Time: 03:00 hrs.

Full Marks: 60 /Pass Marks: 24

MCA103/MCA113: Operating System (*New/Old Course*)

Candidates are required to give their answers in their own words as far as practicable.

Group A

Answer TWO questions.

2×12=24

1. Explain deadlock with its different conditions. Explain deadlock detection and prevention techniques. 4+4+4
2. Why page replacement is needed? Calculate the number of page faults for the given reference string using FIFO and Optimal Page Replacement algorithms with 3 page frames. 2+10
Reference string: 2, 3, 7, 3, 5, 2, 6, 7, 4, 3, 2, 5, 4, 5
3. What is process scheduling and what are the performance criteria to evaluate process scheduling? Explain preemptive and non-preemptive SJF process scheduling with suitable example. 4+8

Group B

Answer SIX questions.

6×6=36

4. Discuss operating system with its types and functions.
5. Define safe and unsafe states. State Banker's algorithm with single resource.
6. Explain different types of fragmentation with examples.
7. What is file? Discuss different operations performed on files.
8. Discuss different types of terminals in brief.
9. What are different types of disk scheduling algorithm? Explain FCFS disk scheduling with suitable example.
10. What is distributed operating system? Discuss the advantages of distributed system over centralized system.
11. Write short notes on any TWO:
(a) Principles of I/O software (b) DMA (c) Segmentation

